

When descriptive forecast-horizon gains do not imply global evidence of incremental meteorological predictability: a multi-site PM₁₀ study

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Abstract

Improvements over persistence are often reported as extended forecast skill at longer horizons, but such descriptive gains need not imply correspondingly strong statistical support that an added information set contributes incremental predictive signal. We evaluate hourly **PM₁₀** forecasts at nine sites (Madrid Casa de Campo and eight Irish stations) over January–July 2023 using an expanding-window rolling-origin scheme with 24-h spacing between forecast origins and horizons $h = 1, \dots, 24$. The incremental comparison pairs direct multi-horizon XGBoost models using pollutant lags with the same learner augmented by meteorological observations available at the forecast origin (an observational information-set experiment rather than the use of future numerical weather prediction (NWP) forecasts). In Madrid, meteorology yields a pronounced descriptive change in the skill profile, increasing the strict longest-contiguous positive persistence-relative skill horizon from 9 to 17 h (+8 h), while the relaxed last-positive-skill horizon increases from 15 to 17 h (+2 h). Yet in 36 dimensionally corrected comparisons using the Diebold–Mariano test with Harvey–Leybourne–Newbold correction (DM-HLN) at $h \in \{1, 6, 12, 24\}$, no test survives global Bonferroni correction under either the overlap-based specification ($q_{\text{overlap}} = 0$; 0/36) or a fixed automatic Newey–West sensitivity allowing positive-lag dependence (0/36). The evidence therefore supports a clear descriptive horizon extension at some sites, but no site–horizon comparison survives global multiplicity control under either audited dependence specification; only limited within-station signals remain.

Keywords: PM₁₀, rolling-origin evaluation, persistence-relative skill, forecast horizon, Diebold–Mariano test, multiplicity

1 Introduction

Forecast usefulness in air quality is inherently horizon-dependent: a model that improves near-term accuracy may offer little advantage later, and the practical value of an hourly PM₁₀ forecast depends on how long it remains better than a simple persistence reference. Persistence is therefore more than a baseline convenience: it sets

a demanding yardstick for whether a forecasting system adds information beyond the most recent observation, in a setting where concentrations are shaped by emissions and meteorologically driven transport and mixing (Seinfeld and Pandis 2006; Kumar et al. 2016).

Reported improvements from meteorological covariates complicate interpretation because pollutant history already reflects some of the same processes. When meteorology is added to a data-driven model, an observed reduction in error may represent genuinely incremental information, or it may reflect a re-expression of structure already present in the lagged PM₁₀ signal, with the balance varying by site and by lead time. Accordingly, performance gains documented across the literature span a wide range of designs and outcomes (Méndez et al. 2023; Pak et al. 2018; Bai et al. 2018; Ansari and Quaff 2025).

This motivates an interpretive gap that is often left implicit: extending the range of horizons over which a model outperforms persistence is descriptively meaningful, but it does not by itself resolve how strongly that extension can be attributed to the added meteorological information once sampling uncertainty and multiple comparisons are taken into account.

We study this issue in a nine-site hourly PM₁₀ experiment (Madrid Casa de Campo and an eight-station Irish network) evaluated over January–July 2023 with an expanding-window rolling-origin design. Persistence and SARIMA provide reference benchmarks, and the incremental comparison pairs direct multi-horizon XGBoost models based on pollutant lags with the same learner augmented by meteorological observations available at the forecast origin. Horizon-wise performance is summarised descriptively through persistence-relative RMSE skill curves and horizon descriptors H^* , while inferential attribution is assessed with DM-HLN tests (Diebold and Mariano 1995; Harvey et al. 1997) at $h \in \{1, 6, 12, 24\}$ under global multiplicity control across the 36 planned site–horizon comparisons.

Against this design, the results show that sizeable horizon extensions can arise in a way that looks compelling descriptively, yet provide weak family-wise global evidence for incremental meteorological predictability once the planned inferential family is accounted for. At the same time, the family-wise conclusion is unchanged when residual positive-lag dependence is allowed through a fixed automatic Newey–West sensitivity, clarifying which aspects of the evidence are global and stable and which remain local and specification-sensitive.

2 Data

2.1 Madrid

Hourly PM₁₀ concentrations and concurrent meteorological measurements were obtained for Madrid Casa de Campo from the Madrid City Council open-data portal. The station is classified as an urban-background site on the western edge of the city, away from major point sources, and is representative of citywide background conditions. The study period spans January 2020 through July 2023; records with missing PM₁₀ or meteorological data were flagged and excluded from model training and evaluation on a per-origin basis. Seven meteorological variables were retained: precipitation (*rain*), temperature (*temp*), relative humidity (*rhum*), mean sea-level pressure (*msl*), wind speed (*wdsp*), wind direction (*wddir*), and solar radiation.

2.2 Ireland

Hourly PM₁₀ concentrations and meteorological covariates were obtained from the Irish Environmental Protection Agency (EPA) monitoring network. Eight stations distributed across Ireland were selected on the basis of data completeness over the study window (January 2020 – July 2023): Birr (Co. Offaly), Dublin Airport, Dundalk (Co. Louth), Pearse Street Dublin, Ringsend Dublin, Edenderry (Co. Offaly), Henry Street Limerick, and Portlaoise (Co. Laois). Meteorological variables were extracted from the nearest mapped EPA synoptic station for each monitoring site.

Table 1 summarises the basic PM₁₀ statistics for all nine sites over the training and evaluation periods. Concentrations are generally low to moderate, with training-period means ranging from 11.3 $\mu\text{g m}^{-3}$ (Portlaoise) to 19.1 $\mu\text{g m}^{-3}$ (Dublin Airport), and Madrid at 18.4 $\mu\text{g m}^{-3}$. The training-period 95th percentile exceeds 40 $\mu\text{g m}^{-3}$ at four of nine sites: Madrid, Dublin Airport, Ringsend Dublin, and Edenderry. Data completeness is high: missing rates are below 2% at all sites during both periods.

Table 1 Descriptive statistics of hourly PM₁₀ ($\mu\text{g m}^{-3}$) for all nine sites. Training: 2020–2022; evaluation: Jan–Jul 2023. n : valid hourly observations; miss: missing rate (%).

Station	Training (2020–2022)				Evaluation (2023)			
	n	Mean	SD	P95	n	Mean	SD	P95
Madrid	25 630	18.4	20.0	42.0	4 988	15.9	12.3	34.6
Birr	20 372	12.8	12.5	31.4	5 085	13.2	11.3	31.2
Dublin Airport	21 958	19.1	16.1	46.9	3 504	15.7	10.9	34.3
Dundalk	21 938	11.4	8.6	24.3	3 658	13.1	10.2	30.7
Pearse St. Dublin	17 009	13.0	9.9	30.2	5 088	15.4	10.1	33.3
Ringsend Dublin	21 949	15.6	16.1	41.9	5 087	14.0	13.1	36.7
Edenderry	11 526	17.8	19.8	48.8	5 029	15.9	16.0	38.5
Henry St. Limerick	13 128	13.0	12.1	30.8	5 052	11.9	8.6	27.2
Portlaoise	21 958	11.3	9.7	27.5	5 088	10.9	7.9	25.1

3 Methods

3.1 Backtesting protocol

We adopt a rolling-origin expanding-window backtesting scheme to preserve prospective temporal information flow and strict separation between training and evaluation data. For each forecast origin t , the training window covers all available observations up to (but not including) t ; no future information is used at any point. The forecast origin advances in daily strides of 24 h, producing one $h = 1, \dots, 24$ forecast vector per day.

The evaluation period was 1 January–31 July 2023, with forecast origins spaced by 24 h. The number of valid evaluated origins varied by site according to data availability. The mandatory minimum training window is 8 760 observations (one year), so origins in early 2023 draw on three years of history. This protocol prevents look-ahead from future observations in the rolling-origin evaluation. Model configurations and feature sets used in the evaluation were held fixed across evaluation origins; no origin-level tuning or cross-validation was performed.

3.2 Model specifications

3.2.1 Persistence.

The persistence model predicts $\hat{y}_{t+h} = y_t$ for all h . It serves as the denominator benchmark for the skill score.

SARIMA.

We fit a SARIMA(1,0,1)(1,0,0)₂₄ model at each origin using the *statsmodels* implementation of exact maximum likelihood. To avoid numerical instability of the Kalman filter on very long training windows, the training series is truncated to the most recent 17 520 observations (approximately two calendar years) at each origin. Multi-step-ahead forecasts are generated by iterating the fitted state-space model forward. SARIMA is treated as a univariate statistical reference; it does not receive meteorological inputs.

XGBoost — lags only.

A separate XGBoost regressor (Chen and Guestrin 2016) is trained for each horizon $h \in \{1, \dots, 24\}$ (the *direct* multi-horizon strategy). The feature set consists of eight lagged PM₁₀ values ($y_{t-1}, y_{t-2}, y_{t-3}, y_{t-6}, y_{t-12}, y_{t-24}, y_{t-48}, y_{t-168}$) and four calendar indicators (hour of day, day of week, month, Julian day). Hyperparameters are fixed across all origins and horizons (Table 2) to ensure fair comparison; no origin-level cross-validation is performed.

XGBoost — lags + meteorology.

Identical to the lags-only model except that the meteorological variables measured at time t are appended to the feature vector (seven for Madrid, nine for the Irish stations). This is an *observational* setup in which meteorological information is available at the moment of forecast

Table 2 Fixed XGBoost hyperparameters used for all models and horizons.

Parameter	Value
Number of estimators	300
Maximum tree depth	4
Learning rate	0.05
Row subsampling	0.90
Column subsampling	0.90
Objective	reg:squarederror
Random seed	42

issuance. Madrid uses concurrent station observations, whereas the Irish covariates come from the nearest mapped EPA synoptic station. Because observed meteorological values are used rather than numerical weather prediction (NWP) forecasts, the reported skill scores should not be interpreted as operational performance with NWP inputs (see Section 5).

3.3 Evaluation metrics

3.3.1 Skill score.

For each model m and horizon h , the RMSE skill score relative to persistence is:

$$S_m(h) = 1 - \frac{\text{RMSE}_m(h)}{\text{RMSE}_{\text{pers}}(h)}, \quad (1)$$

where the RMSEs are computed over all forecast origins in the evaluation period. Positive $S_m(h)$ indicates that model m outperforms persistence at horizon h .

Useful forecast horizon H^* .

We report two descriptive persistence-relative horizon summaries based on the sign of $S_m(h)$ over $h \in \{1, \dots, 24\}$:

- H_{strict}^* (**primary**): the length of the longest *contiguous* run of horizons with $S_m(h) > 0$.
- H_{relax}^* (**secondary**): the last horizon h for which $S_m(h) > 0$.

H^* is a descriptor of the skill-curve sign pattern and is not itself a hypothesis test.

Diebold–Mariano test (DM-HLN), dependence specification, and multiplicity.

Incremental meteorological information is assessed inferentially via horizon-wise Diebold–Mariano tests (Diebold and Mariano 1995) with the Harvey–Leybourne–Newbold finite-sample correction (Harvey et al. 1997) (DM-HLN), comparing squared-error losses of XGBoost (lags only) versus XGBoost (lags + meteorology) at physical horizons $h \in \{1, 6, 12, 24\}$.

Because forecasts are issued once per day (24-h origin spacing), the loss-differential sequence advances in daily steps rather than in physical hours. We therefore define $h_{\text{DM}} = \lceil h/24 \rceil$ as the horizon expressed in loss-differential steps. For $h \in \{1, 6, 12, 24\}$, $h_{\text{DM}} = 1$ for all tests, so the mechanically induced overlap bandwidth is $q_{\text{overlap}} = \max(0, h_{\text{DM}} - 1) = 0$. This overlap-based choice removes the *mechanically required* positive-lag covariance from overlapping multi-step forecast errors, but it does not imply that loss differentials are independent or that residual dependence cannot exist.

To address potential positive-lag dependence not induced mechanically by overlap, we also report a fixed automatic Newey–West sensitivity analysis (Newey and West 1987) using the bandwidth rule $q_{\text{auto}} = \lfloor 4(n/100)^{2/9} \rfloor$.

The primary inferential family comprises the 36 planned site–horizon comparisons (9 sites $\times$ 4 horizons), controlled with global Bonferroni correction. As a within-station sensitivity analysis addressing a narrower inferential scope, we also apply Benjamini–Hochberg false-discovery-rate control separately within each station (4 horizons per station).

4 Results

4.1 Madrid—substantial descriptive horizon extension

Madrid offers the clearest visual contrast between the two XGBoost information sets. Figure 1 shows that the lags-only model is persistence-superior only over a limited portion of the 24-h range, whereas the meteorology-augmented model maintains positive persistence-relative skill further into the day. Summarising this pattern with the primary strict descriptor, the longest-contiguous

positive-skill run increases from $H_{\text{strict}}^* = 9$ h (lags only) to $H_{\text{strict}}^* = 17$ h (lags + meteorology), a strict extension of +8 h. The relaxed descriptor highlights a more modest change at the tail: H_{relax}^* increases from 15 to 17 h (+2 h). Reporting both values is useful because they summarise different aspects of the same skill profile—continuity versus the last isolated horizon with $S(h) > 0$ —and, in turn, frame how compelling the descriptive horizon gain appears before any formal attribution.

4.2 Cross-site heterogeneity and limited observable headroom

The Irish network presents a different descriptive geometry. In many panels of Figure 2, persistence-relative skill remains positive for much of $h = 1, \dots, 24$ under the lags-only specification, so a strict horizon descriptor often reaches the design boundary even before meteorology is added. Consistent with this limited headroom, the mean strict horizon increases from 21.875 to 22.875 h (mean increment +1.000 h), and five of eight stations already attain $H_{\text{strict}}^* = 24$ h with lags only. Across both information sets, 11 station/condition cases reach $H_{\text{strict}}^* = 24$ h, so several strict horizons are right-censored by $H_{\text{max}} = 24$ rather than indicating a natural predictability limit.

Table 3 provides the corresponding station-level horizon summaries. Taken together, the Madrid and Ireland profiles suggest that “hours gained” is not a stable descriptive object across sites, motivating a shift from visual horizon extensions to inferential support for incremental meteorological information.

4.3 Descriptive gains do not scale to family-wise global support

The paired loss-differential comparisons provide that inferential layer. Under the primary dimensionally coherent overlap-based DM-HLN specification ($q_{\text{overlap}} = 0$), all 36 planned station-horizon tests are valid (Table 4). While three comparisons yield raw $p < 0.05$, none survives the global Bonferroni correction across the 36-test family (0/36). At a narrower scope, station-wise Benjamini–Hochberg control retains three local signals (Birr at $h = 1$, Madrid at $h = 12$, and Pearse St. Dublin at $h = 6$). These outcomes are compatible: local signals can coexist with the

absence of family-wise rejection, and 0/36 under the global correction should not be interpreted as equivalence or proof of no effect.

4.4 Global conclusion remains unchanged under fixed automatic Newey–West sensitivity

The overlap-based choice $q_{\text{overlap}} = 0$ addresses dependence mechanically induced by forecast overlap under 24-h origin spacing, but it does not preclude residual positive-lag dependence in the loss differentials. Table 5 therefore reports a fixed automatic Newey–West sensitivity. The global picture is unchanged: again, no comparison survives global Bonferroni correction (0/36). Three within-station Benjamini–Hochberg signals remain, but their identities are not identical. Birr at $h = 1$ and Pearse St. Dublin at $h = 6$ are retained under both specifications; Madrid at $h = 12$ appears only under the overlap-based specification, whereas Pearse St. Dublin at $h = 12$ appears under the automatic Newey–West sensitivity.

The evidence thus separates cleanly by scale: the family-wise global conclusion is stable across the two audited dependence specifications, while the limited local signals are more sensitive to that choice.

5 Discussion

The descriptive evidence initially points in different directions depending on the site. In Madrid, adding meteorological observations reshapes the horizon-wise skill profile enough to extend the strict positive-skill run from 9 to 17 h, a change that is hard to ignore when viewed against persistence. Yet the relaxed summary moves from 15 to 17 h, reminding us that even within a single site the apparent magnitude of a “horizon extension” depends on which descriptive question H^* is asked to answer.

The Irish network adds a second layer of qualification. Many lags-only skill curves remain positive throughout most of the evaluated range, so the strict descriptor often reaches $H_{\text{max}} = 24$ before meteorology is introduced. The mean strict horizon still increases, but only from 21.875 to 22.875 h (+1.000 h), and with five of eight

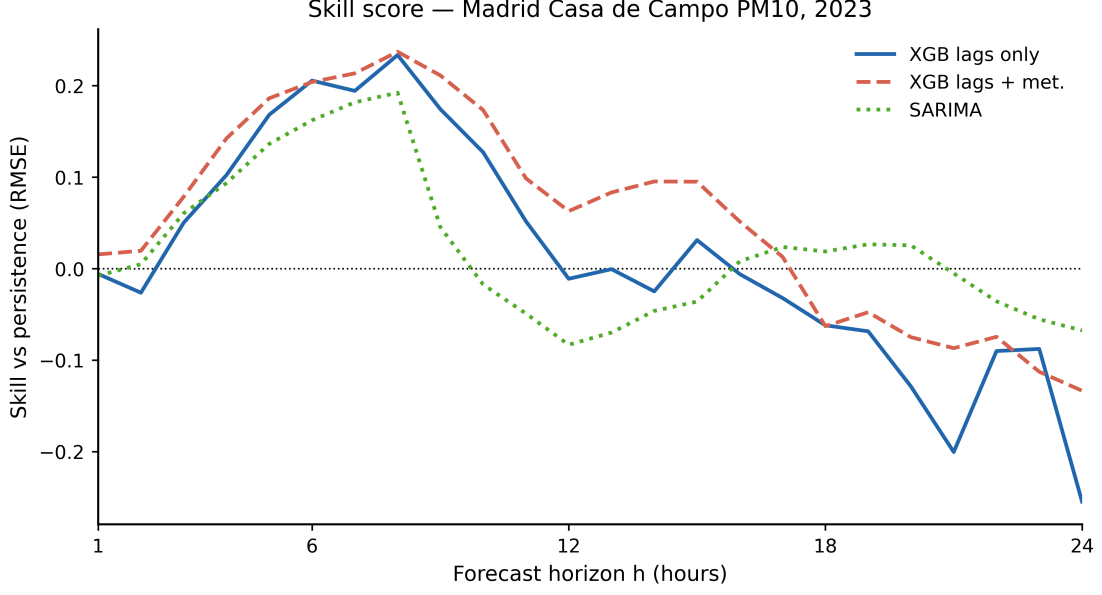


Fig. 1 Madrid Casa de Campo (Jan–Jul 2023): persistence-relative RMSE skill $S_m(h) = 1 - \text{RMSE}_m(h)/\text{RMSE}_{\text{pers}}(h)$ over physical horizons $h = 1, \dots, 24$ h. Positive values indicate lower RMSE than persistence. Blue solid: XGBoost (lags only). Red dashed: XGBoost (lags + meteorology, using meteorological observations available at the forecast origin). Green dotted: SARIMA(1,0,1)(1,0,0)₂₄. Horizontal line: $S = 0$.

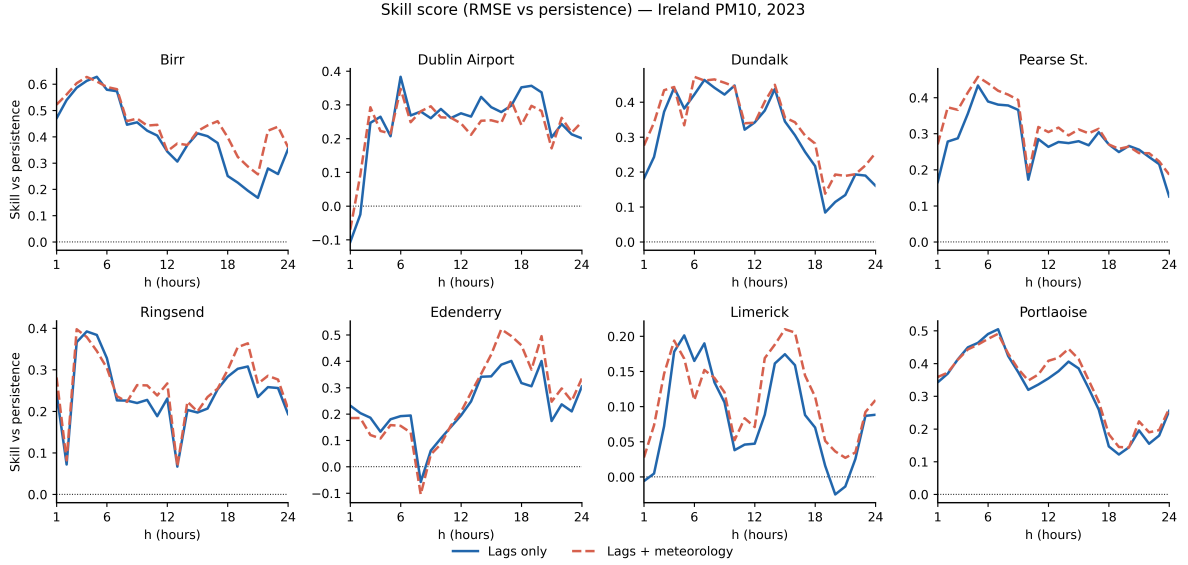


Fig. 2 Ireland (Jan–Jul 2023): persistence-relative RMSE skill $S_m(h) = 1 - \text{RMSE}_m(h)/\text{RMSE}_{\text{pers}}(h)$ over physical horizons $h = 1, \dots, 24$ h for eight stations (one panel per station). Blue solid: XGBoost (lags only). Red dashed: XGBoost (lags + meteorology, using meteorological observations available at the forecast origin). Horizontal line: $S = 0$. Stations with $H_{\text{strict}}^* = 24$ reach the 24-h evaluation boundary.

lags-only cases already at the boundary (11 sta-
tion/condition cases at $H_{\text{strict}}^* = 24$ in total). In

this setting, small observed changes in H^* are con-
sistent with limited observable headroom rather

than with a definitive absence of incremental information.

Once the comparison is moved from descriptive summaries to paired loss-differential inference, the scale of the evidential claim becomes explicit. Under the primary corrected DM-HLN specification ($q_{\text{overlap}} = 0$), none of the 36 planned tests survives global Bonferroni correction, even though a small number of within-station Benjamini–Hochberg signals remain. The absence of family-wise rejection should not be read as equivalence; rather, it bounds what can be supported as a multi-site, multi-horizon attribution of the descriptive gains to the added meteorological information.

The dependence specification is the most immediate statistical pressure point for this conclusion. Because $q_{\text{overlap}} = 0$ removes only dependence mechanically induced by overlapping multi-step forecast errors, residual serial dependence in the loss-differential sequence remains a plausible concern. The fixed automatic Newey–West sensitivity addresses that concern without tuning to outcomes. It leaves the family-wise global result unchanged (again 0/36 after global Bonferroni), while the identity of the limited local signals shifts across specifications, tempering any strong claim of local stability.

The evaluation geometry also carries a methodological lesson: when forecast origins are spaced differently from physical horizons, DM-HLN/HAC parameters must be expressed in loss-differential sequence steps rather than transferred mechanically from hours.

Within these bounds, the study supports a restrained interpretation. Meteorological covariates are origin-time observations rather than future NWP forecasts; the 24-h horizon limit censors H^* for several stations; and the evidence is specific to the nine sites, the January–July 2023 evaluation window, and the evaluated XGBoost design. The automatic Newey–West analysis is a fixed sensitivity rather than an exhaustive dependence model, and descriptive improvements do not provide causal attribution. Taken together, the results argue for keeping descriptive horizon extension, local inferential signals, and family-wise global evidence as distinct levels of support when interpreting incremental information in environmental forecasting.

6 Conclusions

The evidence in this nine-site PM_{10} study is easiest to read when descriptive horizon summaries and inferential attribution are kept separate. Descriptively, the Madrid skill profile changes markedly when meteorological observations are added, extending the strict longest-contiguous positive-skill horizon from 9 to 17 h (with a smaller relaxed change from 15 to 17 h), while the Irish network exhibits heterogeneous and frequently boundary-limited behaviour within $H_{\text{max}} = 24$.

When the same lags-only versus lags+meteorology comparison is assessed through the 36 planned, dimensionally corrected DM-HLN tests, no comparison survives global Bonferroni correction under either the overlap-based specification ($q_{\text{overlap}} = 0$) or the fixed automatic Newey–West sensitivity. The global conclusion is therefore stable to this bounded dependence treatment, even though the identities of the limited local BH signals are not.

For multi-horizon environmental forecasting studies, the implication is bounded: a compelling extension of persistence-relative skill can be descriptively substantial, yet still fall short of supporting a family-wise global attribution to incremental meteorological predictability under the planned inferential family.

Table 3 Ireland station-level H^* (hours) by model and criterion. H_{strict}^* is the primary longest-contiguous positive-skill horizon; H_{relax}^* is the secondary last-positive-skill horizon. Entries of 24 for H_{strict}^* indicate reaching the 24-h evaluation boundary ($H_{\text{max}} = 24$).

Station	XGBoost (lags only)		XGBoost (lags + met.)		SARIMA	
	Strict	Relax	Strict	Relax	Strict	Relax
Birr (Co. Offaly)	24	24	24	24	16	24
Dublin Airport	22	24	23	24	24	24
Dundalk (Co. Louth)	24	24	24	24	18	24
Pearse St. Dublin	24	24	24	24	17	23
Ringsend Dublin	24	24	24	24	9	20
Edenderry (Co. Offaly)	16	24	16	24	9	21
Henry St. Limerick	17	24	24	24	4	17
Portlaoise (Co. Laois)	24	24	24	24	17	17
Mean	21.875	24.0	22.875	24.0	14.3	21.3
Mean $\Delta H_{\text{strict}}^*$		+1.000 h			—	

Table 4 Primary DM-HLN inference under the dimensionally coherent overlap-based specification. Forecast origins are spaced by 24 h, so for physical horizons $h \in \{1, 6, 12, 24\}$ the loss-differential horizon in sequence steps is $h_{\text{DM}} = \lceil h/24 \rceil = 1$ and the mechanically induced overlap bandwidth is $q_{\text{overlap}} = \max(0, h_{\text{DM}} - 1) = 0$ for all tests. $q_{\text{overlap}} = 0$ removes mechanically required overlap dependence but does not imply independence. Global Bonferroni is applied across the 36 planned site-horizon tests; Benjamini-Hochberg (BH) is applied within each station (4 horizons) as a local sensitivity analysis.

Specification	Value
Planned comparisons (sites $\times$ horizons)	$9 \times 4 = 36$
Valid tests	36/36
Raw $p < 0.05$	3
Global Bonferroni significant	0/36
Station-wise BH significant (count)	3
Station-wise BH identities	Birr ($h = 1$); Madrid ($h = 12$); Pearse St. Dublin ($h = 6$)

Table 5 Fixed automatic Newey-West sensitivity analysis for DM-HLN inference. The HAC bandwidth is set by $q_{\text{auto}} = \lfloor 4(n/100)^{2/9} \rfloor$, allowing positive-lag autocovariances in the loss-differential sequence. Global Bonferroni is applied across the 36 planned tests; BH is applied within each station (4 horizons) as a local sensitivity analysis.

Specification	Value
Planned comparisons (sites $\times$ horizons)	$9 \times 4 = 36$
Valid tests	36/36
Raw $p < 0.05$	5
Global Bonferroni significant	0/36
Station-wise BH significant (count)	3
Station-wise BH identities	Birr ($h = 1$); Pearse St. Dublin ($h = 6$); Pearse St. Dublin ($h = 12$)

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Statements and Declarations

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Competing interests

The author has no relevant financial or non-financial interests to disclose.

Author contributions

The author conceived the study, designed the methodology, performed the analyses, developed the software, interpreted the results, and wrote and approved the manuscript.

Data availability

The public GitHub repository https://github.com/fedeg-umh-es/P1_PM10_Meteorology_Hstar contains the versioned analysis code, the processed and regenerated study artifacts archived there, figure-generation support, run metadata, and the final overlap-based and automatic Newey–West inferential tables reported in this manuscript for the underlying meteorology experiment. The Irish station evaluations were regenerated from recovered source datasets using the versioned analysis pipeline. The row-level predictions from the originally executed Irish computational run were not retained; therefore, the reported Irish results correspond to the documented regeneration rather than recovery of the original prediction files. Raw source data are available from the Madrid City Council and the Irish Environmental Protection Agency under their respective open-data terms.

Ethics approval and consent to participate

Not applicable. The study used environmental monitoring data and involved no human participants or animals.

Consent for publication

Not applicable.

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